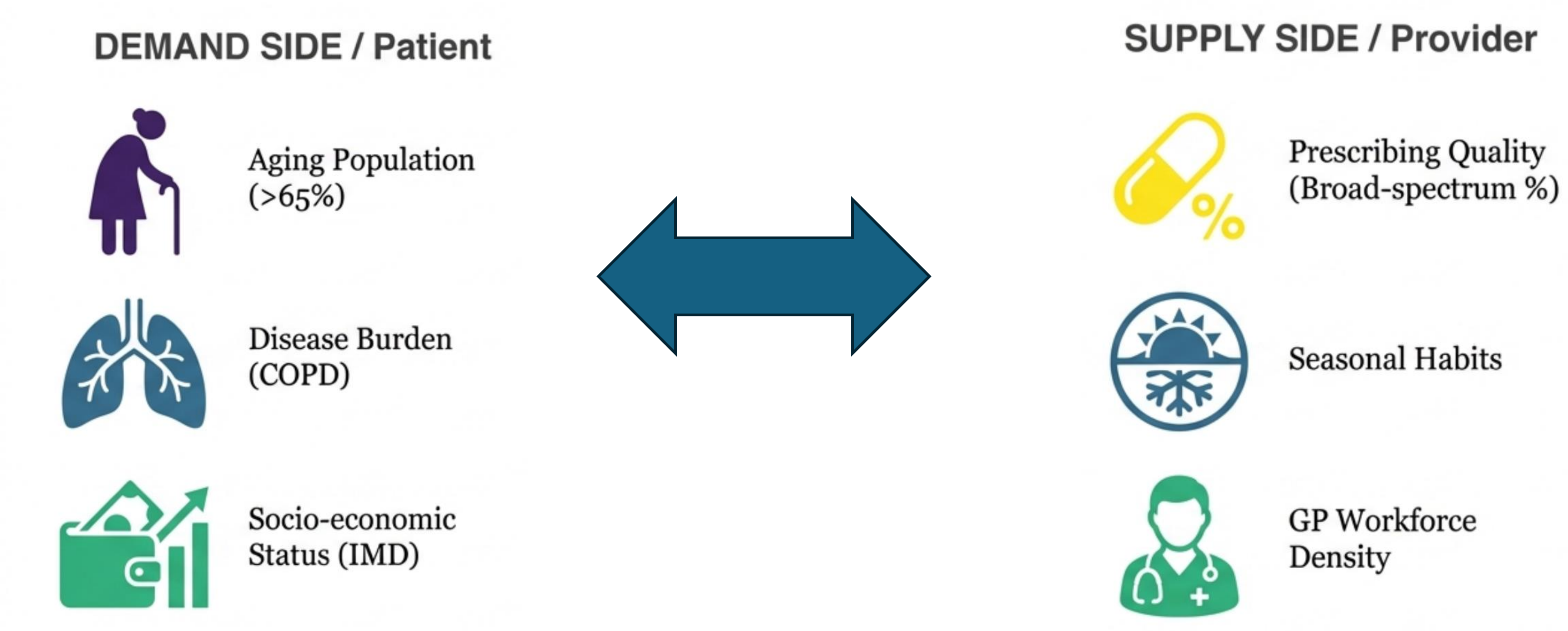
Regional Antibiotic Prescribing Patterns in England: A Business Analytics approach to Healthcare:

Clustering & Explainable AI Analysis

Analysis of 106 Sub-ICBs (2024-2025) | Data Source: Fingertips API

Antibiotics as a Product: Beyond the Prescribed Medicine



Assumption: regional consumption outcomes are the result of the friction between clinical demand and health system supply.

What We Don’t Know About the Antibiotic Market

Hot spots of antibiotic prescribing appear in the East, South and West of London.

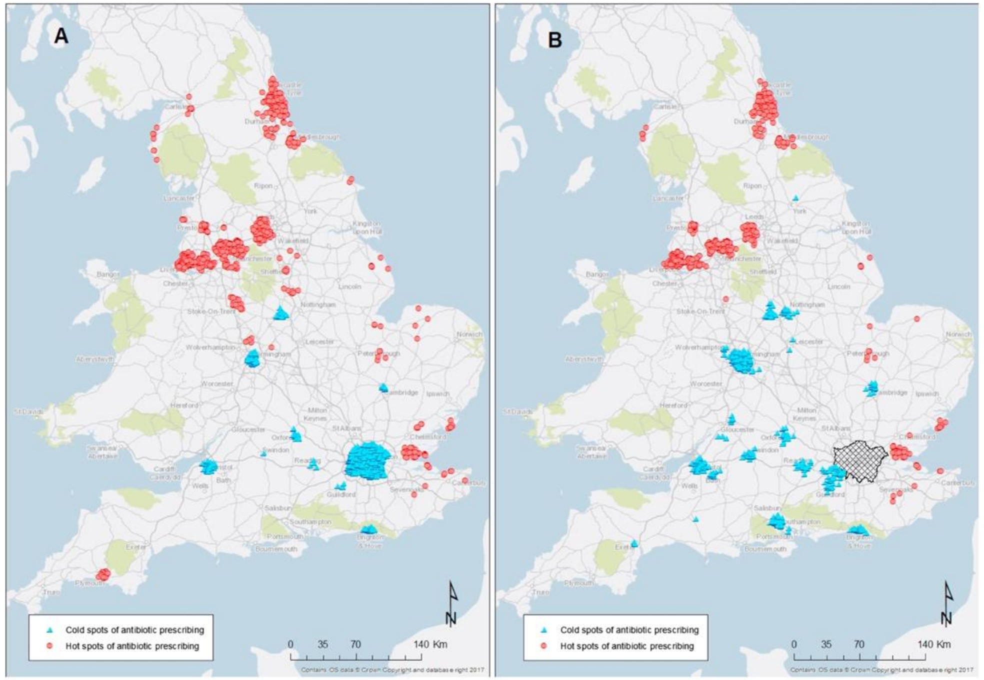


Fig. 1. Hot and Cold spots of antibiotic prescribing in English GP practices in 2016. A: All GP. B: Excluding GP in the London CCGs.

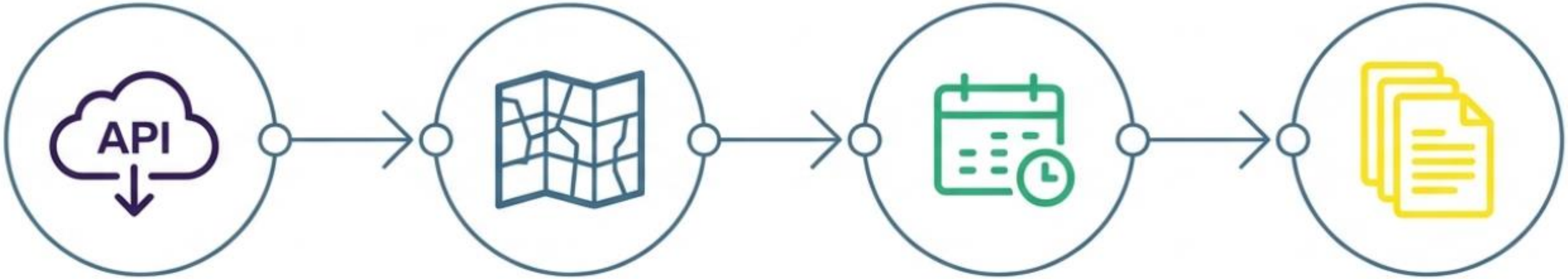
Mölter, M. Belmonte, V. Palin, et al.Antibiotic prescribing patterns in general medical practices in England: does area matter?Health Place, 53 (2018), pp. 10-16

Current Strategy: One Size Fits All

Reality: Distinct Market Archetypes

1 Do distinct regional archetypes exist?
2. What factors (supply vs. demand) drive them?
3. Do drivers operate differently across archetypes?

Data Acquisition &Scope



Source:
Fingertips API

Scope:
106 Sub-ICBs

Treating 106 Sub-ICBs as 106 distinct regional markets

Timeframe:
2024/2025

Metrics:
Prescribing data merged with social determinants (GP density, IMD, COPD)

Characterizing Regional 'Product Consumption' Profiles

Feature	What It Measures	Business Interpretation
Intensity	Items per 1,000 patients	Market volume
Quality	Broad-spectrum antibiotic %	Product mix (refined vs crude)
Seasonality	Winter/summer ratio	Demand stability
Trend	Year-over-year change %	Market growth/decline

The Analytics Pipeline: Segmentation &Attribution

Stage 1: Market Segmentation
(Unsupervised Learning)

Method: K-Means Clustering on 4-dimensional feature data.

Goal: Group 106 regions based on purely behavioral patterns.

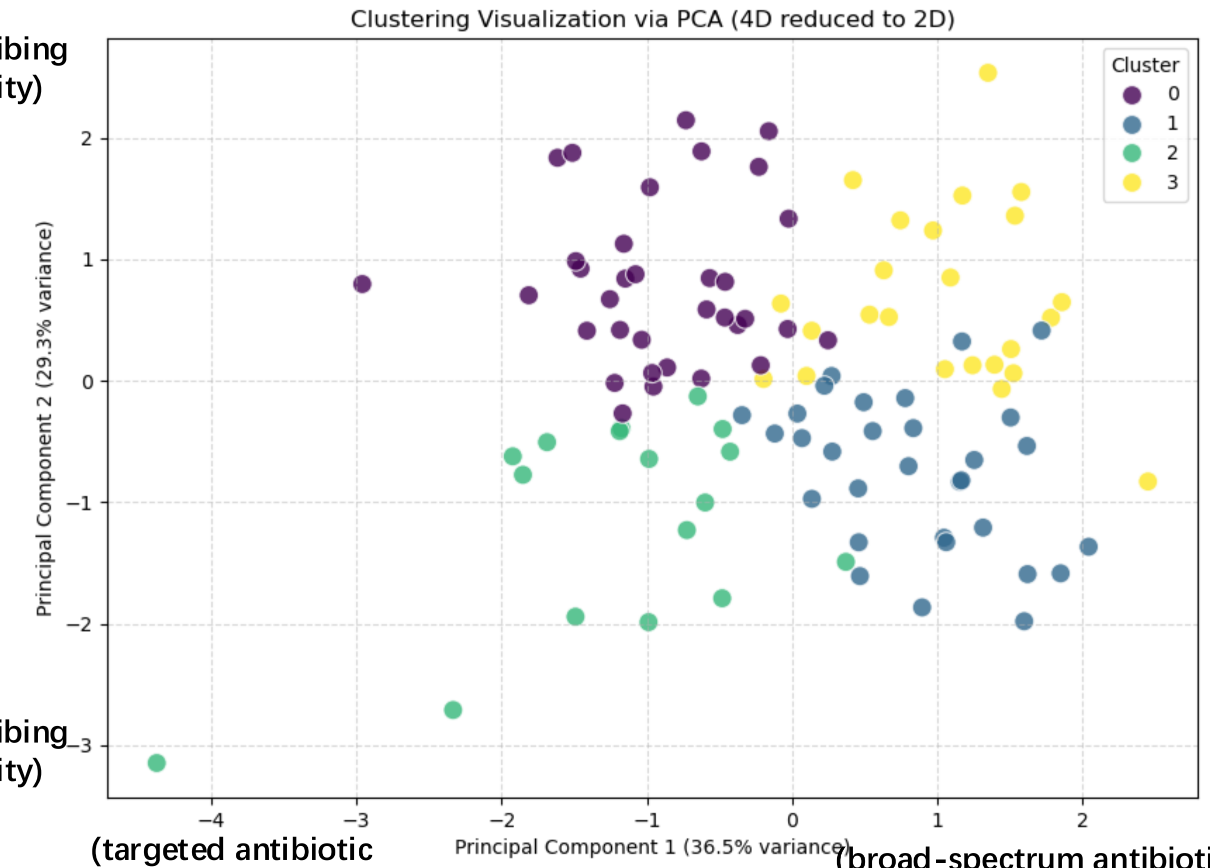
Stage 2: Driver Attribution
(Explainable AI)

Method: Random Forest Model + SHAP

Goal: Quantify exactly why a region behaves the way it does (Accuracy R2 ~ 0.94).

Four Distinct Regional Market Archetypes

(High Prescribing Intensity)



(Low Prescribing Intensity)

(targeted antibiotic stable)

(broad-spectrum antibiotic Seasonal)

(High Prescribing Intensity)

(targeted antibiotic stable)

(broad-spectrum;Seasonal)

(Low Prescribing Intensity)

Cluster 0:

High intensity
Targeted prescribing

high antibiotic demand but well-managed with low broad-spectrum usage.

Cluster 3:

High intensity
High broad-spectrum, seasonal prescribing

Great demand and complex management

Cluster 2:

Low intensity
Targeted prescribing

Low overall prescribing and refined prescribing.

Cluster 1:

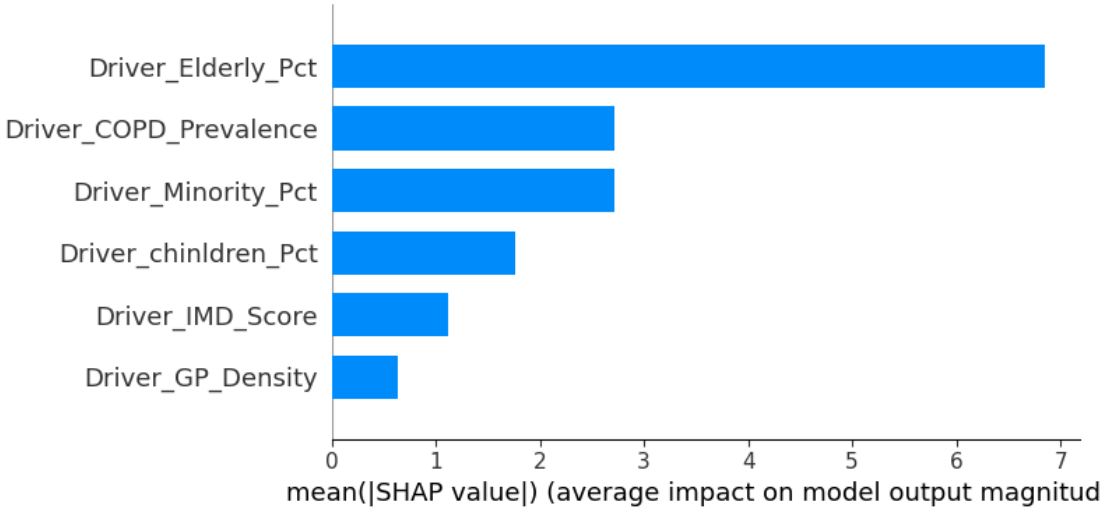
Low intensity
High broad-spectrum seasonal prescribing

Low overall prescribing but high tendency towards broad-spectrum prescribing and seasonal fluctuation.

K-means clustering algorithm grouped 106 regions into 4 distinct types

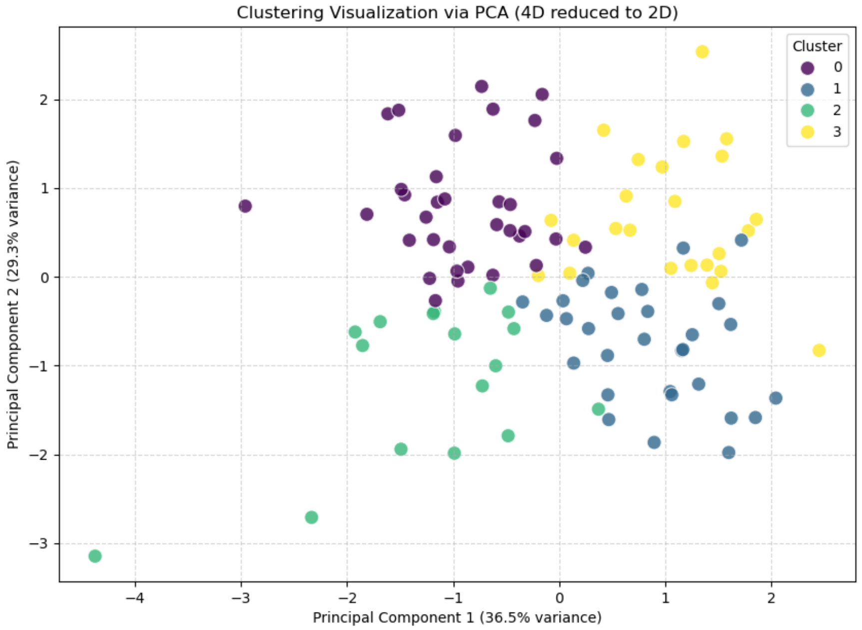
Why National Averages Miss the Story

National Level Analysis

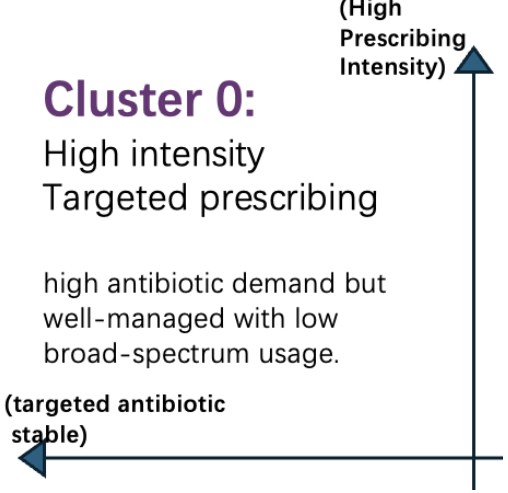


Nationally, the Elderly population (>65) appears to be the only dominant driver. Other factors look negligible.

Segmented Reality



However, this average masks the complex interplay of deprivation, workforce density, and ethnicity specific to each sub-segment.

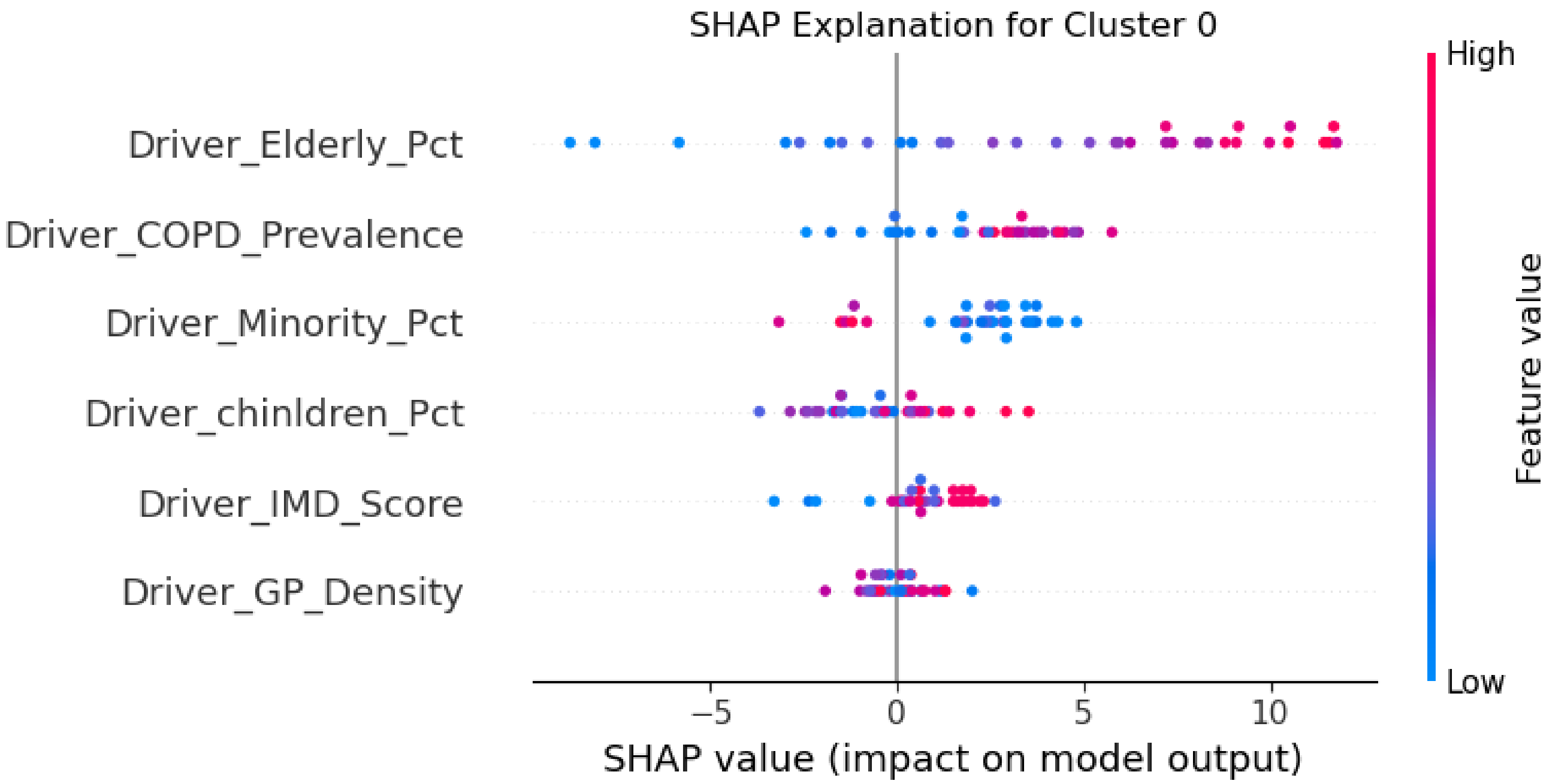


Cluster 0: High Volume, High Efficiency

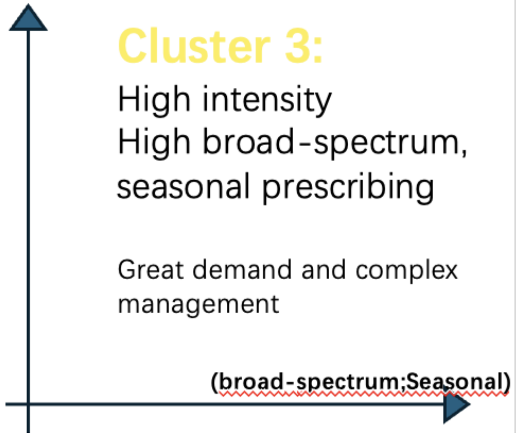
26 Regions | High Intensity, Targeted Prescribing

Drivers:

- Demand: Aging Population is the main driver. COPD prevalence is a secondary, but still significant, positive driver.
- Supply: Efficient management despite high volume.



Insight:
These systems are stressed but working. They have high genuine clinical need (elderly) but manage it with narrow-spectrum precision

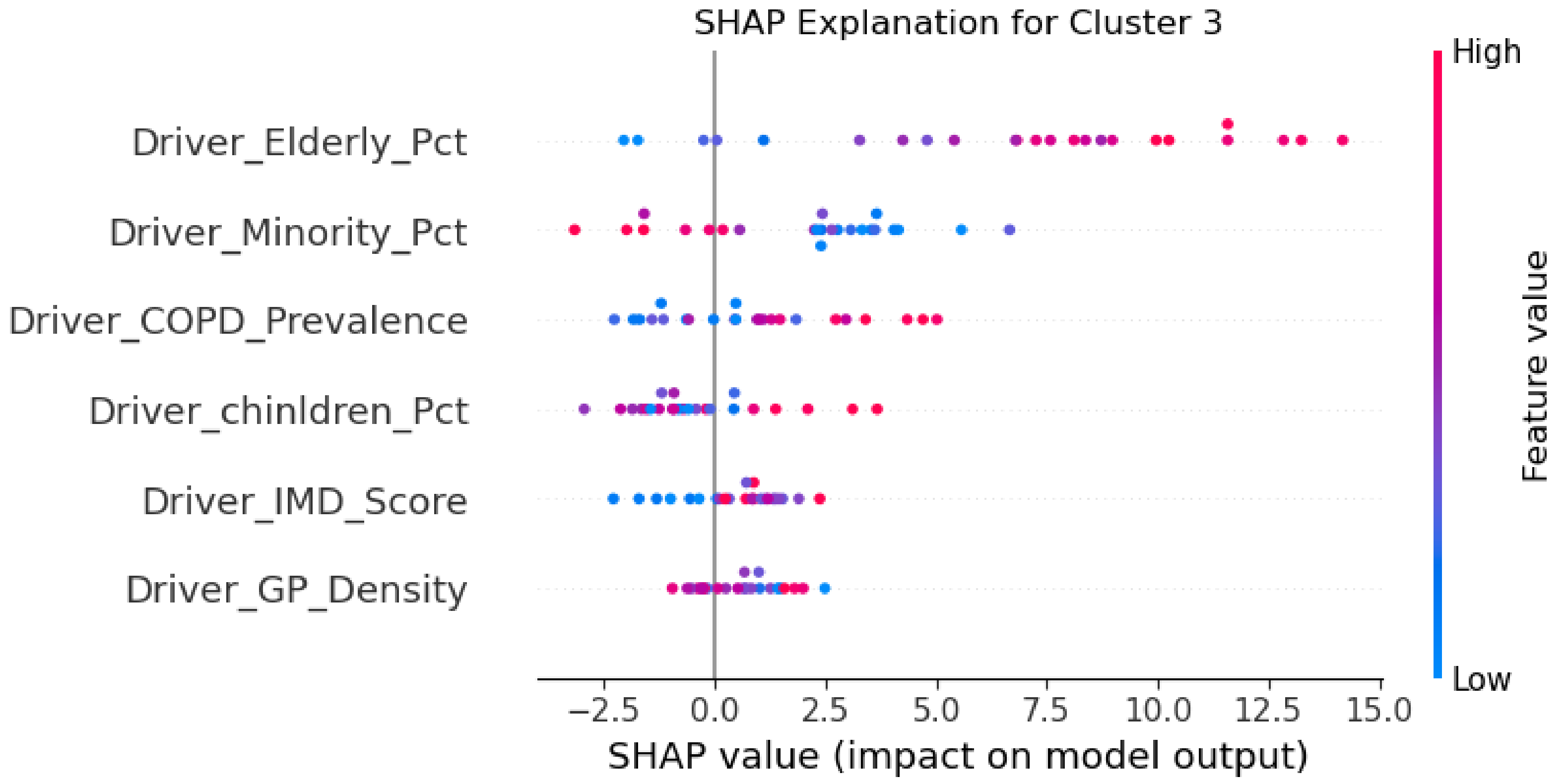


Cluster 3: The 'Double Burden' (High Risk)

27 Regions | High Intensity, Broad-Spectrum, Highly Seasonal

Drivers:
Demand: Aging population +Social complexity.
Supply: Low GP Density correlates significantly with higher prescribing.

Insight:
High clinical demand meets a shortage of GP supply. This suggests 'defensive prescribing'-using broad-spectrum antibiotics due to lack of time for precise diagnosis.



Intervention: Increase workforce density to allow time for better stewardship.

Cluster 2:

Low intensity
Targeted prescribing

Low overall prescribing and refined prescribing.

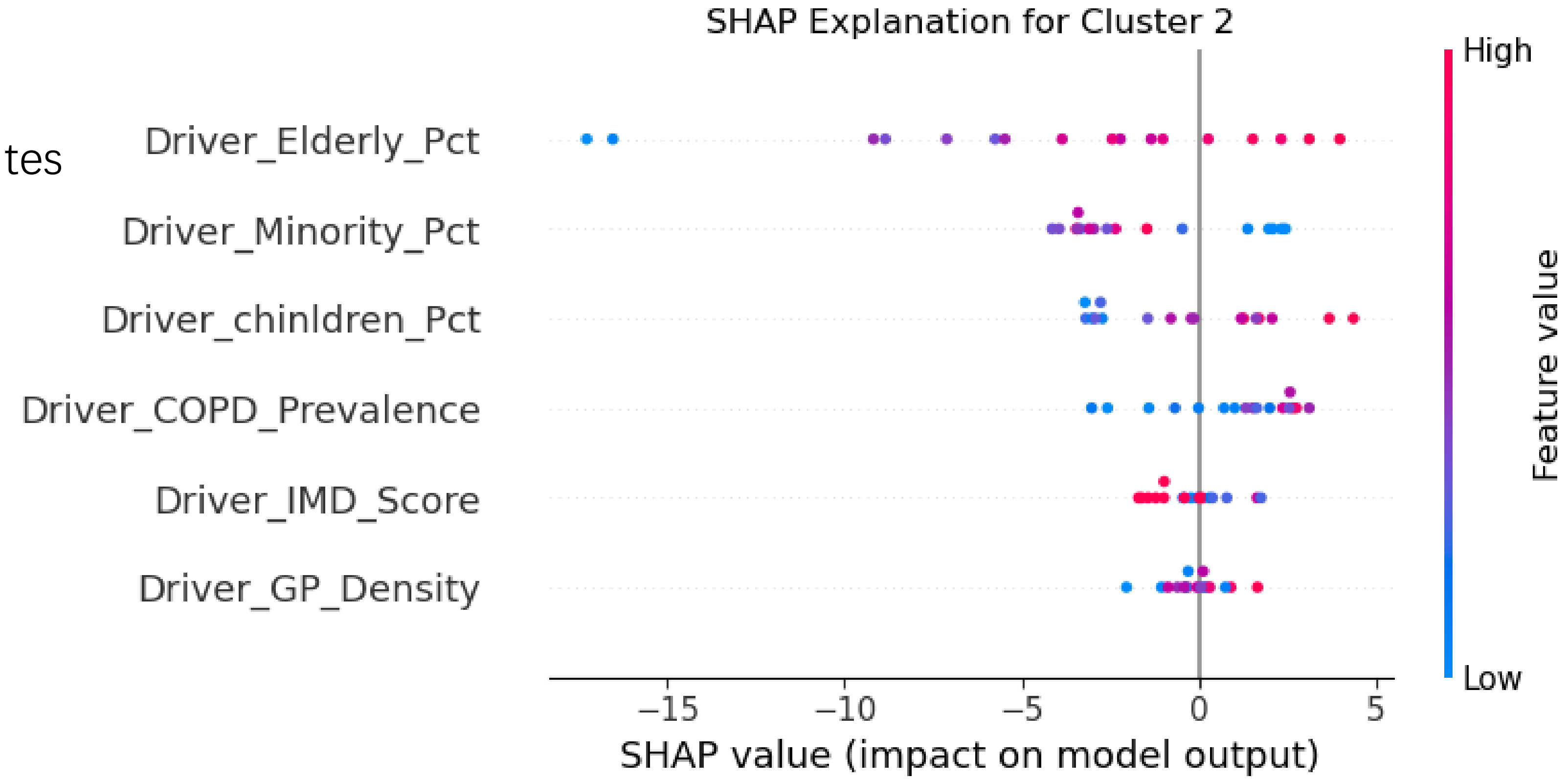
Cluster 2: The Safe Havens? Or an Access Paradox?

29 Regions | Low Intensity, Targeted

Drivers:
Demand: Low elderly population.
The Paradox: High IMD (deprivation) correlates with lower' antibiotic use
in this cluster.

Insight:
Low prescribing here might not be 'good stewardship' but rather 'poor access'
due to poverty. The high deprivation/low usage correlation is a red flag.

Intervention:
Investigate the reason under low volume.
Ensure the low volume isn't hiding unmet clinical needs.



(broad-spectrum;Seasonal)

Cluster 1:
Low intensity
High broad-spectrum
seasonal prescribing

Low overall prescribing but high tendency towards broad-spectrum prescribing and seasonal fluctuation.

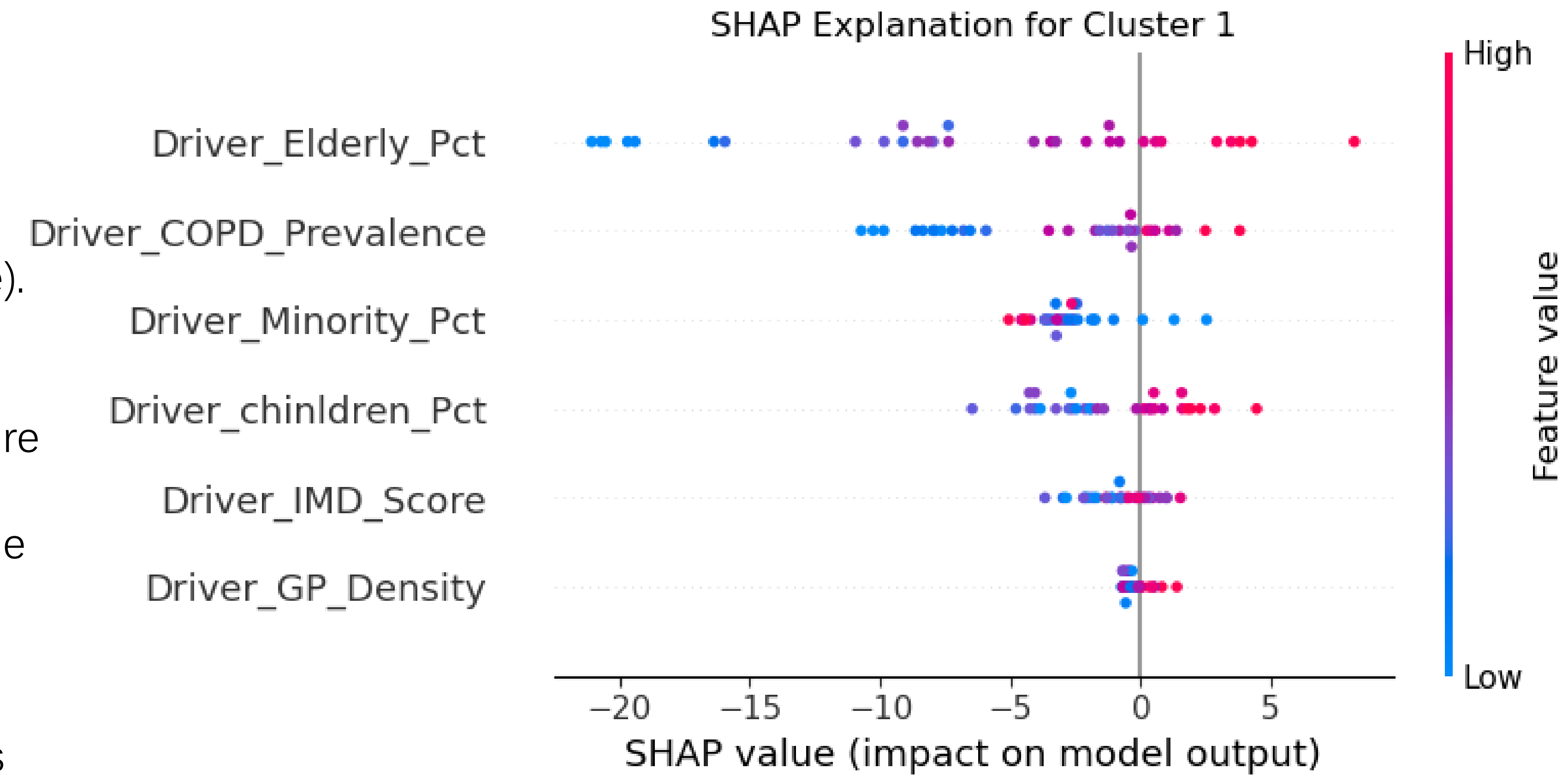
Cluster 1: Inefficient Potential

24 Regions Low Intensity, but Broad-Spectrum/Seasonal

Drivers:
- Demand: Young population (keeps overall volume low).
-Supply: Poor prescribing habits (high tendency toward broad-spectrum use).

Insight:
Volume is low only because patients are healthy. If this population ages, this cluster will immediately into the double burden category.

intervention:
Need educational interventions. Focus on prescribing guidelines and stewardship training.



Summary:

Regional distinct archetypes exist.

Each cluster faces different drivers.

Differentiated interventions is necessary.

Limitations & Future Focus

Limitation:

- **Ecological Fallacy: Sub-ICB data represents regional averages, not individual patient behaviors.**
- **Cross-sectional data analysis (2024-25) cannot prove causation, only correlation.**
- **Context: Does not capture all local factors , does not consider clinical indicators / context.**

Future Direction:

- **Longitudinal time-series analysis.**
- **Deep into individual GP Practice level (include more clinical context).**

Thank you for listening !